The Pie diagram is so called because the entire graph looks like an American food 'pie' and the components like slices cut from 'pie'.


American food 'pie'

### 6.3.2 Method of constructing a pie chart:

In a pie chart, we know that the various components are represented by the sectors of a circle and the whole circle represents the sum of the value of all the components. Therefore, the total angle of 360° at the centre of the circle is divided into different sectors according to the value of the components.

$$\textbf{The central angle of a component} = \frac{value\ of\ the\ component}{total\ value} \times 360°$$

Sometimes, the value of the components are expressed in percentage. In such cases,

$$\textbf{The central angle of a component} = \frac{percentage\ value\ of\ the\ component}{100} \times 360°$$

**Steps for construction of the pie chart:**

1) Calculate the central angle for each component using the above formula and tabulate it.
2) Draw a circle of convenient radius and mark one horizontal radius in it.
3) Draw radius making central angle of first component with horizontal radius. This sector represents the first component. From this radius, draw next radius with central angle of the second component and so on, until the completion of all components.
4) For identification of each sector, shade with different colours.
5) Label each sector.

Here are given some examples, let us draw the pie chart for the given data.

**Example 6.4**

Draw a pie diagram to represent the following data, which shows the expenditure of paddy cultivation in 2 acres of land.

| **Particulars** | Seeds | Ploughing | Wages | Fertilizer | Harvest | Others |
|---|---|---|---|---|---|---|
| **Expenses (₹)** | 2000 | 6000 | 10000 | 7000 | 8000 | 3000 |

Also, 1. Find the percentage of the head in which more money had been spent?

2. What percentage of money was spent for seeds?



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